

Camera Calibration Project Report

Computer Vision and Pattern Recognition

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M.Invernici

R.Ortese

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Abstract

This report comes from a collaboration and merges two solutions of the same exam project. We calibrate a pinhole camera using Zhang’s method [1] with a planar checkerboard: we detect corners, estimate one homography per image, recover the intrinsic matrix K , compute extrinsics (R_k, t_k) , validate via reprojection error and augmented-reality superimposition, study how the number of calibration images affects the stability of the estimated intrinsics, and ultimately compare the estimated R with the provided poses.

0 Introduction

Camera calibration [2] estimates camera intrinsic and extrinsic through the mapping between 3D world points and 2D image measurements. Under the pinhole model, a 3D point $X = (X, Y, Z, 1)^\top$ projects to an image point $x = (u, v, 1)^\top$ as

$$s x = P X, \quad P = K [R \mid t], \quad (1)$$

where K contains intrinsic parameters (focal lengths in pixels, principal point, skew), (R, t) are the extrinsics for each view and s is an arbitrary nonzero scale factor (in practice proportional to the point depth Z_c in the camera frame).

Zhang’s method exploits a planar calibration pattern: for a plane $Z = 0$, the mapping from planar coordinates $X_p = (X, Y, 1)^\top$ to the image is a homography H :

$$s x = H X_p, \quad H = K [r_1 \ r_2 \ t]. \quad (2)$$

By estimating multiple homographies from different views and enforcing the orthonormality constraints of (r_1, r_2) , we can recover K in closed

form, then compute (R_k, t_k) for each image. Calibration quality is assessed through reprojection error and visual checks.

Dataset note. In our run, chessboard corners were successfully detected in **81** images (the calibration is carried out on this subset).

0.1 Load images and detect chessboard corners

Each image is converted to grayscale since corner detection relies on intensity gradients. We detect the internal checkerboard intersections using OpenCV’s `findChessboardCorners` (with optional refinement via `cornerSubPix`). Only images where detection succeeds are kept; for each accepted image we store:

- the ordered 2D corner coordinates $\{(u_i, v_i)\}_{i=1}^N$ in pixel units,
- the corresponding object (world) points $\{(X_i, Y_i, 0)\}_{i=1}^N$ on the plane.

0.2 World coordinates on the checkerboard plane

We build a grid of planar points aligned with the checkerboard, scaled by the physical square size s (in meters or millimeters):

$$X_i = i \cdot s, \quad Y_i = j \cdot s, \quad Z_i = 0. \quad (3)$$

Each corner corresponds to a planar homogeneous image coordinate $X_p^{(i)} = (X_i, Y_i, 1)^\top$.

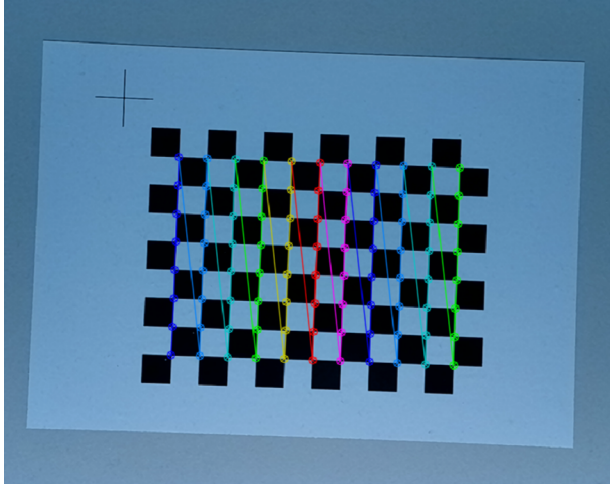


Figure 1: Example of detected checkerboard corners.

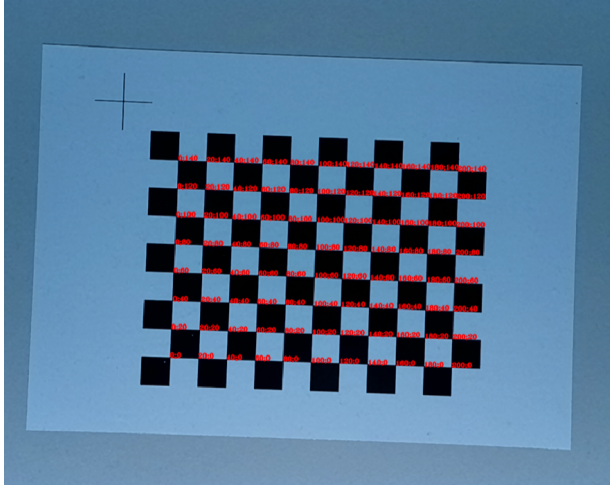


Figure 2: Real world coordinates of detected checkerboard corners from sample image.

1 Zhang's method

1.1 Homography estimation via DLT

For each calibration image k , we estimate the homography $H^{(k)}$ from planar world points to image points using the Direct Linear Transform (DLT) [2]. Each correspondence contributes two linear equations, producing a homogeneous system

$$Ah = 0, \quad A \in \mathbb{R}^{2N \times 9}, \quad h = \text{vec}(H). \quad (4)$$

Because measurements are noisy, we solve using SVD:

$$A = U\Sigma V^\top, \quad (5)$$

where h is the right singular vector associated with the smallest singular value. Finally, we

normalize the scale (e.g., $H_{33} = 1$).

1.2 Estimate the intrinsic matrix K from homographies

Let h_1, h_2, h_3 be the columns of a homography H . Since

$$H \sim K [r_1 \ r_2 \ t], \quad (6)$$

and r_1, r_2 are orthonormal, we obtain the constraints:

$$r_1^\top r_2 = 0 \Rightarrow h_1^\top B h_2 = 0, \quad (7)$$

$$r_1^\top r_1 = r_2^\top r_2 \Rightarrow h_1^\top B h_1 = h_2^\top B h_2, \quad (8)$$

with $B = (K K^\top)^{-1}$ symmetric.

Each constraint can be written linearly in the 6-vector b containing the independent entries of B . Using the standard v_{ij} construction,

$$h_i^\top B h_j = v_{ij}^\top b, \quad v_{ij} = \begin{bmatrix} h_{1i}h_{1j} \\ h_{1i}h_{2j} + h_{2i}h_{1j} \\ h_{2i}h_{2j} \\ h_{3i}h_{1j} + h_{1i}h_{3j} \\ h_{3i}h_{2j} + h_{2i}h_{3j} \\ h_{3i}h_{3j} \end{bmatrix}. \quad (9)$$

Thus each homography provides two equations:

$$v_{12}^\top b = 0, \quad (v_{11} - v_{22})^\top b = 0, \quad (10)$$

stacked into a linear system $Vb = 0$ solved via SVD. From b we reconstruct B , then recover K (e.g., via Cholesky / matrix factorization).

1.2.1 Two equivalent implementations (direct vs. vech/duplication)

The two project versions implemented this step in slightly different linear-algebra form:

- **Direct v_{ij} stacking (standard Zhang derivation).** This is the textbook approach shown above.
- **vech approach (half-vectorization).** One notebook explicitly built a duplication matrix to map between $\text{vech}(B)$ and $\text{vec}(B)$ so the constraints can be expressed compactly with symmetric structure.

Outcome: both approaches were successful and led to consistent estimates of K (differences only at numerical precision level).

1.2.2 Intrinsic matrix and interpretation

The intrinsic matrix can be written as

$$K = \begin{bmatrix} fk_u & fk_u \cot(\theta) & u_0 \\ 0 & fk_v / \sin(\theta) & v_0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (11)$$

so only the pixel-domain focal lengths (α_x, α_y) and the principal point (u_0, v_0) are directly observable; f, k_u, k_v are not individually identifiable.

Estimated K (example run). From the merged work, one completed run produced:

$$K \approx \begin{bmatrix} 1009.9997 & 1.6880 & 630.3507 \\ 0 & 1010.6048 & 373.9010 \\ 0 & 0 & 1 \end{bmatrix}. \quad (12)$$

This suggests nearly square pixels ($\alpha_x \approx \alpha_y \approx 1010$ px), small skew (near zero), and a principal point close to the image center.

1.3 Estimate extrinsics (R_k, t_k)

Given $H^{(k)}$ and K , we compute:

$$\begin{aligned} \lambda_k &= \frac{1}{\|K^{-1}h_1^{(k)}\|}, & r_1^{(k)} &= \lambda_k K^{-1}h_1^{(k)}, \\ r_2^{(k)} &= \lambda_k K^{-1}h_2^{(k)}, & t^{(k)} &= \lambda_k K^{-1}h_3^{(k)}. \end{aligned} \quad (13)$$

We set $r_3^{(k)} = r_1^{(k)} \times r_2^{(k)}$.

Orthonormality correction. Due to noise, the raw rotation matrix may not be perfectly orthogonal. We project it to the closest rotation matrix via SVD:

$$R_{\text{raw}}^{(k)} = U\Sigma V^\top, \quad R^{(k)} = UV^\top. \quad (14)$$

Positive depth consistency check. The extrinsic parameters recovered from a homography are defined up to a sign ambiguity. In particular, if $(R^{(k)}, t^{(k)})$ is a valid solution, then $(-R^{(k)}, -t^{(k)})$ may also satisfy the homography constraints. Although the orthonormality correction ensures that $R^{(k)}$ is a proper rotation matrix (with $\det(R^{(k)}) = 1$), it does not guarantee that the reconstructed scene lies in front of the camera.

To ensure physical consistency, we verify that the calibration plane is located in front of the

camera by checking the depth of its points in the camera reference frame:

$$X_c = R^{(k)}X_w + t^{(k)}, \quad Z_c > 0.$$

Since the depth is a linear function over the planar chessboard surface, it is sufficient to evaluate the four outer corners of the board (defined with $Z = 0$ in the world frame). If all corner depths are positive, the entire plane lies in front of the camera and the pose is physically valid.

In our case, the minimum depth across all views is strictly positive, confirming that no sign inconsistency is present and that the estimated extrinsic parameters are geometrically consistent.

2 Compute reprojection error

Calibration parameters are validated by reprojection error: we reproject each known grid point into the image and compare it to the detected corner. Using $P^{(k)} = K[R^{(k)} | t^{(k)}]$ and homogeneous world point $m^{(i)} = (X_i, Y_i, 0, 1)^\top$,

$$\hat{u}^{(i)} = \frac{p_1^\top m^{(i)}}{p_3^\top m^{(i)}}, \quad \hat{v}^{(i)} = \frac{p_2^\top m^{(i)}}{p_3^\top m^{(i)}}, \quad (15)$$

and the total squared reprojection error for one image is

$$\varepsilon(P) = \sum_{i=1}^N \left(\hat{u}^{(i)} - u^{(i)} \right)^2 + \left(\hat{v}^{(i)} - v^{(i)} \right)^2. \quad (16)$$

Example value. For one calibration image, an RMS reprojection error of about **0.305 px** was obtained, indicating accurate calibration.

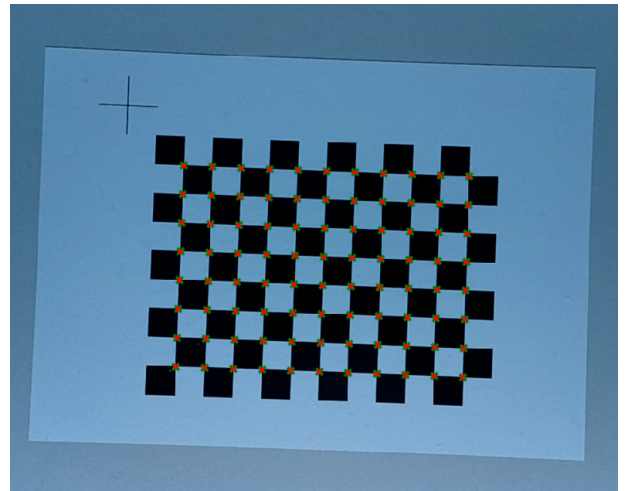


Figure 3: Detected corners (green cross) vs. reprojected points (red dot) for a sample image.

3 Virtual object superimposition

To qualitatively validate calibration, we project a simple 3D object (e.g., a cylinder, circular and elliptical bases) defined in the checkerboard reference frame. With $P^{(k)} = K [R^{(k)} | t^{(k)}]$, any 3D point on the object is mapped to image coordinates and drawn on top of the image. A correct calibration yields:

- the object base attached to the board plane ($Z = 0$),
- consistent perspective across different viewpoints,
- straight projected edges following projective geometry.

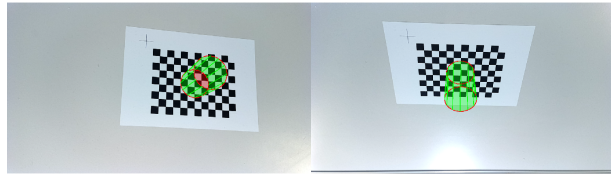


Figure 4: Augmented reality check: cylinder projected on multiple calibration images.

4 Influence of the number of calibration images on K stability

We study the stability of intrinsic parameters as a function of the number of images. Starting from all available homographies, we repeatedly sample random subsets of size k and recompute K using Zhang’s method. For each k we compute the standard deviations of key parameters across subsets, e.g.

$$\begin{aligned} u_0 &= K_{13}, & v_0 &= K_{23}, \\ \alpha_x &= K_{11}, & \alpha_y &= K_{22}. \end{aligned} \quad (17)$$

Empirically, the variability is very large for small k (e.g. $k = 2, 3$), and decreases quickly as k grows; from about $k \approx 6$ images onward

the estimates become stable, meaning additional images mainly improve robustness rather than causing large parameter shifts.

For completeness, the same stability analysis is performed on the focal length f , averaging the results over multiple random seeds to reduce the effect of subset sampling variability.

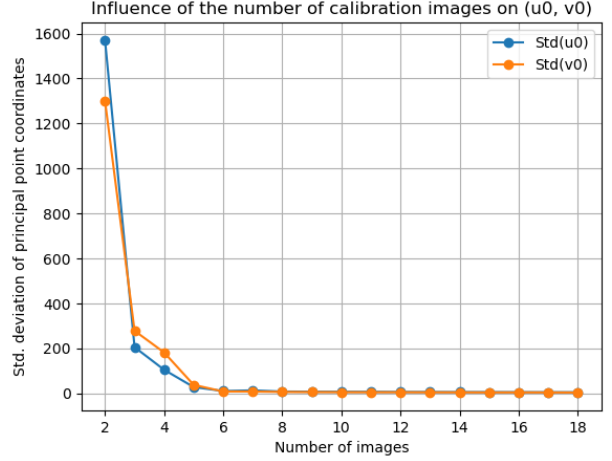


Figure 5: Standard deviation of (u_0, v_0) vs. number of images k .

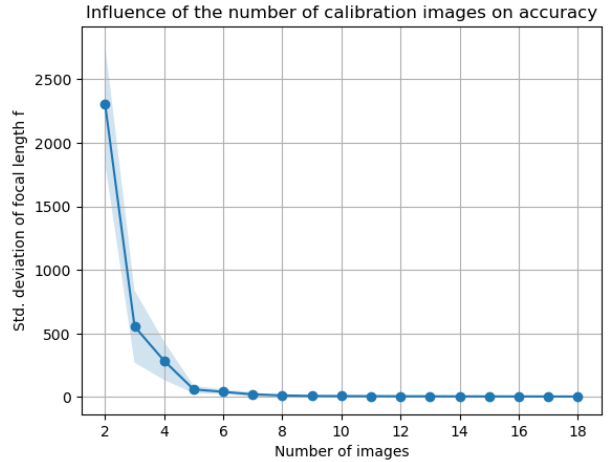


Figure 6: Standard deviation of focal length vs. k (optionally averaged across random seeds).

5 Extrinsic comparison with provided poses

We compared the estimated extrinsics with poses stored in external YAML files (fields `R_CS` and `T_CS`). A meaningful rotation comparison uses

the relative rotation

$$R_{\text{rel}} = R_{\text{ref}} R_{\text{est}}^{\top}, \quad \theta = \arccos \left(\frac{\text{tr}(R_{\text{rel}}) - 1}{2} \right), \quad (18)$$

where $|\theta|$ measures the angular discrepancy.

An initial comparison yielded an apparent error close to 180° , which was explained by a **different checkerboard reference-frame convention** between the estimation code and the YAML poses (e.g., planar axes swap and/or flip of the normal). By introducing a fixed board-frame rotation S such that $X_{\text{yaml}} = S X_{\text{ours}}$, the rotations relate as

$$R_{\text{est}} = R_{\text{yaml}} S. \quad (19)$$

In one run, a rounded estimate of this mismatch was:

$$S \approx \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad (20)$$

corresponding to a swap $X \leftrightarrow Y$ and a flip $Z \mapsto -Z$. After accounting for this, the angular error dropped close to zero, confirming that the calibration extrinsics were consistent and that the large discrepancy was purely a coordinate-convention issue.

6 Conclusion

We implemented Zhang’s planar calibration pipeline and validated it both quantitatively (low reprojection error) and qualitatively (AR superimposition). Two alternative linear implementations for intrinsic estimation (standard v_{ij} stacking and a `vech`/duplication-matrix formulation) were both successful and consistent. A robustness study confirmed that using only a few images leads to unstable estimates, while around six or more views produce a reliable intrinsic matrix. Finally, an extrinsics comparison against provided poses highlighted the importance of consistent reference-frame conventions when interpreting rotation/translation parameters.

References

- [1] Z. Zhang, *A Flexible New Technique for Camera Calibration*, IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 22, no. 11, pp. 1330–1334, 2000.

- [2] Felice Andrea Pellegrino, *Lecture 3-Camera Calibration*, University of Trieste, 554SM–Fall 2025.

AI Assistance. ChatGPT (GPT-5.2, OpenAI) was used to support code debugging and improve the clarity of the report writing. The final implementation and analysis were reviewed and validated by the authors.